

LaTeX Exam Survival Guide

For Timed Online LaTeX Reproduction Labs

Compiled reference — every construct seen across practice papers

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1 Preamble — Load Everything in the First 30 Seconds

Scan the question once. Every recognizable feature below maps to one package. Load all of them immediately so you never stop mid-answer to add one.

```
\documentclass[11pt]{article}
\usepackage[margin=1in]{geometry} % sane margins
\usepackage{amsmath,amssymb,amsthm} % equations, symbols, theorems
\usepackage{authblk} % multi-author + affiliations
\usepackage[table]{xcolor} % colors + colored table cells
\usepackage{enumitem} % custom list bullets/labels
\usepackage{graphicx} % images
\usepackage{wrapfig} % text-wrapped figures
\usepackage{rotating} % rotated figures
\usepackage{multirow} % vertical cell merging
\usepackage{booktabs} % professional table rules
\usepackage{caption}
\usepackage{subcaption} % grouped subfigures (1a, 1b, ...)
\usepackage{listings} % source-code listings
\usepackage{hyperref} % clickable links -- LOAD LAST
```

Signal word in the question	Package
Multiple authors / affiliations	authblk
Table of contents	(core — just \tableofcontents)
Colors, colored table	xcolor with [table] option
Custom bullets, nested lists	enumitem
Wrapped figure	wrapfig
Rotated figure	rotating
Subfigures (1a, 1b, ...)	subcaption
Multirow / multicolumn table	multirow (multicolumn is core)
Theorem / Proposition / Proof	amsthm
Code listing	listings
Hyperlink	hyperref

2 Document Structure, Title Block & Table of Contents

2.1 Bare skeleton

```
\documentclass{article}
\begin{document}
  ... everything visible goes here ...
\end{document}
```

Nothing outside `\begin{document}... \end{document}` is ever rendered — packages and definitions belong in the *preamble* (above `\begin{document}`).

2.2 Single author

```
\title{The Solar System}
\author{Johannes Kepler}
\date{}
\begin{document}
\maketitle
```

2.3 Multiple authors, each with a numbered affiliation

Use `authblk`: `\author[n]{...}` tags a number, `\affil[n]{...}` defines what that number means.

```
\author[1]{Pierre-Simon Laplace}
\author[2]{Oliver Heaviside}
\affil[1]{Acad\'emie des Sciences, Paris, France}
\affil[2]{Royal Society, London, United Kingdom}
```

Renders as:

Pierre-Simon Laplace¹, Oliver Heaviside²

¹Académie des Sciences, Paris, France ²Royal Society, London, United Kingdom

If `authblk` is unavailable, fake it manually: `\author{Name\textsuperscript{1}}` then place the affiliation as an ordinary `\footnotetext` or a centered line below the title.

2.4 Table of contents

```
\tableofcontents
\newpage
```

`\tableofcontents` auto-collects every `\section` and `\subsection` with its page number. **You must compile twice** — the first pass writes the list to a `.toc` file, the second pass typesets it. On the first compile the TOC will look empty or stale; that is expected, not an error. Starred headings (`\section*{...}`) are *not* numbered and do *not* appear in the TOC unless you add them manually with `\addcontentsline{toc}{section}{Name}`.

3 Sectioning & Cross-References

```
\section{Kepler's Laws of Planetary Motion}
\subsection{First Law: Elliptical Orbits}
\label{sec:firstlaw}
...
As shown in Section~\ref{sec:firstlaw}, ...
```

Command	Use on	Renders as
<code>\label{key}</code>	anything numbered	(invisible, stores the number)
<code>\ref{key}</code>	sections/figures/tables/theorems	bare number, e.g. 2.3
<code>\eqref{key}</code>	equations only	number in parens, e.g. (9)
<code>\pageref{key}</code>	anything labeled	the page number it's on

Naming convention that saves you time: prefix labels by type so you never confuse them — sec:, eq:, fig:, tab:, thm:, lst:. With hyperref loaded, every \ref/\eqref becomes a clickable jump link automatically — no extra syntax needed.

4 Text Formatting & Colors

```
\textbf{bold}    \textit{italic}    \texttt{typewriter}
\underline{underline}    \textsc{small caps}    \emph{emphasis}
```

Renders as: **bold** *italic* typewriter underline SMALL CAPS *emphasis*

4.1 Colors

```
\definecolor{accent}{RGB}{0,70,140}
\textcolor{accent}{colored text}
\colorbox{yellow}{highlighted background}
\textcolor{white}{\colorbox{accent}{white text on a colored box}}
```

Renders as: colored text highlighted background white text on a colored box

Colors can be called by name (red, blue, gray!20 for 20% gray, teal) or defined once with \definecolor{name}{RGB}{r,g,b} and reused everywhere — always define custom colors once in the preamble.

5 Lists — The Complete Menu

5.1 The three base environments

```
\begin{itemize}    ... \item bullet point ... \end{itemize}
\begin{enumerate} ... \item numbered point ... \end{enumerate}
\begin{description} ... \item[Term] explanation ... \end{description}
```

5.2 Custom bullet symbols (needs enumitem)

```
\begin{itemize}[label=$\oplus$] \item ... \end{itemize}
\begin{itemize}[label=$\triangleright$] \item ... \end{itemize}
\begin{itemize}[label=$\diamond$] \item ... \end{itemize}
\begin{itemize}[label=\textdagger] \item ... \end{itemize}
```

Renders as:

⊕ opus bullet

▷ triangleright bullet

◇ diamond bullet

† dagger bullet (text mode, not math)

5.3 Custom enumerate labels: Roman numerals, letters

```
\begin{enumerate}[label=(\Roman*)] \item Inner planets \item Outer planets \end{enumerate}
\begin{enumerate}[label=(\alph*)] \item first \item second \end{enumerate}
```

Renders as:

- (I) Inner planets
- (II) Outer planets
- (a) first
- (b) second

5.4 Nesting: a group heading with a dagger sub-list (the exact exam pattern)

```
\begin{enumerate}[label=(\Roman*)]
  \item Inner planets
    \begin{itemize}[label=\textdagger]
      \item Mercury, Venus, Earth, and Mars belong to this group.
      \item They are comparatively small and have solid surfaces.
    \end{itemize}
\end{enumerate}
```

Renders as:

- (I) Inner planets
 - † Mercury, Venus, Earth, and Mars belong to this group.
 - † They are comparatively small and have solid surfaces.

The rule for any nested list: open the inner `itemize/enumerate` *inside* the outer `\item`, before that item ends. Indentation is only for your own readability — LaTeX ignores it.

6 Math Symbols Mega-Reference

Everything here needs only `amsmath` + `amssymb` (both in the master preamble).

6.1 Greek letters

Name	Code	Symbol
alpha	<code>\alpha</code>	α
beta	<code>\beta</code>	β
gamma	<code>\gamma</code>	γ
delta	<code>\delta</code>	δ
epsilon	<code>\varepsilon</code>	ε
theta	<code>\theta</code>	θ
lambda	<code>\lambda</code>	λ
mu	<code>\mu</code>	μ
nu	<code>\nu</code>	ν
xi	<code>\xi</code>	ξ
pi	<code>\pi</code>	π
varpi	<code>\varpi</code>	ϖ
rho	<code>\rho</code>	ρ
sigma	<code>\sigma</code>	σ
tau	<code>\tau</code>	τ
phi	<code>\phi</code> / <code>\varphi</code>	ϕ / φ
chi	<code>\chi</code>	χ
psi	<code>\psi</code>	ψ
omega	<code>\omega</code>	ω

Name	Code	Symbol
Gamma	<code>\Gamma</code>	Γ
Delta	<code>\Delta</code>	Δ
Theta	<code>\Theta</code>	Θ
Lambda	<code>\Lambda</code>	Λ
Xi	<code>\Xi</code>	Ξ
Pi	<code>\Pi</code>	Π
Sigma	<code>\Sigma</code>	Σ
Phi	<code>\Phi</code>	Φ
Psi	<code>\Psi</code>	Ψ
Omega	<code>\Omega</code>	Ω
<i>Physics-flavored extras</i>		
solar mass sign	<code>\odot</code> (as $M_\odot$)	$M_\odot$
partial	<code>\partial</code>	∂
nabla	<code>\nabla</code>	∇
infinity	<code>\infty</code>	∞

6.2 Relations, arrows & operators

Name	Code	Sym	Name	Code	Sym
geq	<code>\geq</code>	$\geq$	implies (chain)	<code>\implies</code>	$\implies$
leq	<code>\leq</code>	$\leq$	Longrightarrow	<code>\Longrightarrow</code>	$\Longrightarrow$
neq	<code>\neq</code>	$\neq$	Rightarrow	<code>\rightarrow</code>	$\Rightarrow$
approx	<code>\approx</code>	$\approx$	to/rightarrow	<code>\to</code>	$\rightarrow$
propto	<code>\propto</code>	$\propto$	mapsto	<code>\mapsto</code>	$\mapsto$
equiv	<code>\equiv</code>	$\equiv$	xrightarrow (labeled)	<code>\xrightarrow{\mathcal{L}}</code>	$\xrightarrow{\mathcal{L}}$
in	<code>\in</code>	$\in$	sum	<code>\sum</code>	$\sum$
subset	<code>\subset</code>	$\subset$	product	<code>\prod</code>	$\prod$
forall	<code>\forall</code>	$\forall$	integral	<code>\int</code>	$\int$
exists	<code>\exists</code>	$\exists$	contour integral	<code>\oint</code>	$\oint$
emptyset	<code>\emptyset</code>	$\emptyset$	square root	<code>\sqrt{x}</code>	$\sqrt{x}$
pm	<code>\pm</code>	$\pm$	times	<code>\times</code>	$\times$
mp	<code>\mp</code>	$\mp$	cdot	<code>\cdot</code>	$\cdot$

6.3 Decorations & tensor/index notation

Name	Code	Renders as
bar	<code>\bar{x}</code>	$\bar{x}$
hat	<code>\hat{x}</code>	$\hat{x}$
vector	<code>\vec{x}</code>	$\vec{x}$
dot (time derivative)	<code>\dot{x}</code>	$\dot{x}$
double dot	<code>\ddot{x}</code>	$\ddot{x}$
tilde	<code>\tilde{x}</code>	$\tilde{x}$
subscript+superscript	<code>x_i^2</code>	x_i^2
paired lower indices (tensor)	<code>G_{\mu\nu}</code>	$G_{\mu\nu}$
mixed indices (Christoffel)	<code>\Gamma^{\mu}_{\alpha\beta}</code>	$\Gamma^{\mu}_{\alpha\beta}$
differential with index	<code>dx^{\mu}</code>	dx^{μ}
second time-derivative	<code>\dfrac{d^2 x^{\mu}}{d\tau^2}</code>	$\frac{d^2 x^{\mu}}{d\tau^2}$

Key habit: in general-relativity/tensor style notation, subscripts and superscripts just stack directly onto the base symbol — `G_{\mu\nu}` does not need any special package, it is plain `amsmath` subscript syntax with a two-letter (Greek) index.

7 Equations & Multi-line Derivations

7.1 Numbered vs. unnumbered display math

```
\begin{equation}                                % numbered, one line
  E = mc^2
  \label{eq:emc2}
\end{equation}

\[ E = mc^2 \]                                % unnumbered, one line (or equation*)
```

7.2 Multi-step derivation sharing ONE equation number

Use `align` with `\notag` on every line except the last:

```
\begin{align}
  c^2 &= a^2 - b^2 \notag \\
  e   &= \frac{c}{a} \\
  \label{eq:ecc}
\end{align}
```

Renders as:

$$\begin{aligned} c^2 &= a^2 - b^2 \\ e &= \frac{c}{a} \end{aligned}$$

7.3 Chained equalities across several relations (three-way equality)

```
\begin{align}
  \left(\frac{v}{c}\right)^2 &= \frac{GM}{rc^2} = \frac{r_s}{2r} \\
  \label{eq:weakfield}
\end{align}
```

Renders as:

$$\left(\frac{v}{c}\right)^2 = \frac{GM}{rc^2} = \frac{r_s}{2r}$$

Every `=` sign that must line up vertically across several lines needs its own `&` immediately before it; a single line can have as many extra `=` signs after the aligned one as you like without further `&`s.

7.4 Derivation with arrows between steps, one shared number

```
\begin{align}
  RC[sV_C(s)-v_C(0)]+V_C(s) &= V_{\text{in}}(s), \notag \\
  \rightarrow \quad RCsV_C(s)+V_C(s) &= V_{\text{in}}(s), \notag \\
  \rightarrow \quad V_C(s)(RCs+1) &= V_{\text{in}}(s). \\
  \label{eq:rcderiv}
\end{align}
```

7.5 Fractions, cases, and matrices

Code	Renders as
<code>\frac{a}{b}</code>	$\frac{a}{b}$
<code>\dfrac{a}{b}</code> (full-size, even inline)	$\frac{a}{b}$
<code>\binom{n}{k}</code>	$\binom{n}{k}$

```
f(x) =
\begin{cases}
1 & x \geq 0 \\
-1 & x < 0
\end{cases}

\begin{pmatrix} a & b \\ c & d \end{pmatrix} % round brackets
\begin{bmatrix} a & b \\ c & d \end{bmatrix} % square brackets
\begin{vmatrix} a & b \\ c & d \end{vmatrix} % determinant bars
```

Renders as:

$$f(x) = \begin{cases} 1 & x \geq 0 \\ -1 & x < 0 \end{cases} \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

7.6 Auto-sizing delimiters

Plain () do not grow with tall content; `\left( \right)` do:

```
\left(1 - e^{-t/(RC)}\right) % grows to fit the fraction inside
\left[-\frac{1}{s}e^{-st}\right]_0^{\infty} % sized bracket with limits
```

Renders as: $(1 - e^{-t/(RC)})$ $\left[-\frac{1}{s}e^{-st}\right]_0^{\infty}$

8 Figures & Subfigures

8.1 Plain figure with caption and label

```
\begin{figure}[h]
  \centering
  \includegraphics[width=0.4\textwidth]{myimage.png}
  \caption{A simple series RC circuit.}
  \label{fig:rc}
\end{figure}
```

8.2 Wrapped figure (text flows around it)

```
\begin{wrapfigure}{r}{0.45\textwidth} % r = right side, l = left side
  \centering
  \includegraphics[width=0.43\textwidth]{myimage.png}
  \caption{A simple series RC circuit.}
  \label{fig:rc}
\end{wrapfigure}
```

Needs `wrapfig`. Place it *before* the paragraph you want wrapped around it. Keep the image slightly narrower than the box width or you'll get an overfull-box warning.

8.3 Rotated figure

```
\begin{sidewaysfigure} % whole float rotated 90 degrees
  \centering
  \includegraphics[width=0.6\textwidth]{myimage.png}
  \caption{A rotated figure.}
  \label{fig:rotated}
\end{sidewaysfigure}

\rotatebox{90}{\includegraphics{myimage.png}} % rotate just the image, inline
```

Needs rotating for `sidewaysfigure`. `\rotatebox` comes from `graphicx` (already loaded) and is the faster option when you only need to spin the picture itself, not the whole page float.

8.4 Grouped subfigures with row headers — one Figure, many labeled parts

This is the pattern behind “Figure 1: The Sun and the eight planets...” where sub-images are organized into labeled rows (Star / Inner Planets / Outer Planets) but still share one caption and one figure number. Needs `subcaption`.

```
\begin{figure}[h]
  \centering
  \textbf{\large Group A}\par\medskip
  \begin{subfigure}{0.3\textwidth}
    \centering
    \includegraphics[width=\linewidth]{img_a.png}
  \end{subfigure}
\end{figure}
```

```

        \caption{Item A}
        \label{fig:itemA}
    \end{subfigure}
    \bigskip

    \textbf{\large Group B}\par\medskip
    \begin{subfigure}{0.22\textwidth}
        \centering
        \includegraphics[width=\linewidth]{img_b.png}
        \caption{Item B}
        \label{fig:itemB}
    \end{subfigure}\hfill
    \begin{subfigure}{0.22\textwidth}
        \centering
        \includegraphics[width=\linewidth]{img_c.png}
        \caption{Item C}
        \label{fig:itemC}
    \end{subfigure}\hfill
    \begin{subfigure}{0.22\textwidth}
        \centering
        \includegraphics[width=\linewidth]{img_d.png}
        \caption{Item D}
        \label{fig:itemD}
    \end{subfigure}\hfill
    \begin{subfigure}{0.22\textwidth}
        \centering
        \includegraphics[width=\linewidth]{img_e.png}
        \caption{Item E}
        \label{fig:itemE}
    \end{subfigure}

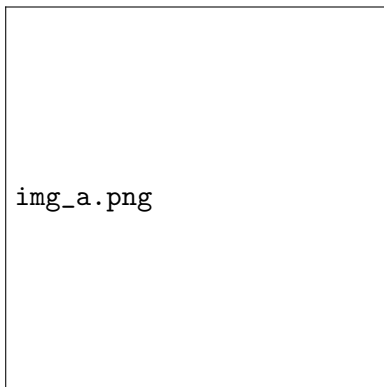
    \caption{Demo figure grouped into two labeled rows.}
    \label{fig:demo}
\end{figure}

```

Renders as:

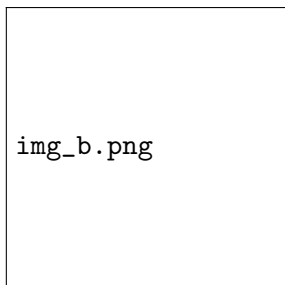
The trick: it is still just *one figure* float. The bold row headers (`\textbf{\large Group A}`) are ordinary text, not part of subcaption’s machinery. Each `subfigure` auto-labels itself (a), (b), (c)... in the order it appears — so referencing `\ref{fig:itemE}` gives **1e** (figure number + letter, e.g. “1e”), exactly like “Figure 1e” in the original. Only the *last* `\caption{}` (outside all the subfigures) creates the overall numbered caption; captions *inside* each `subfigure` create the small (a), (b) sub-labels.

Group A

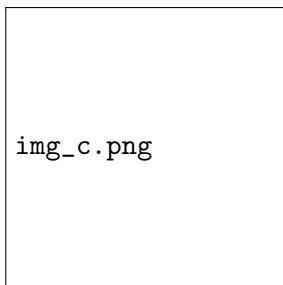


(a) Item A

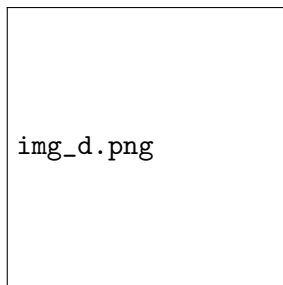
Group B



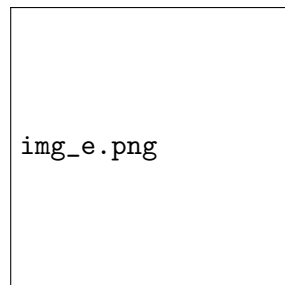
(b) Item B



(c) Item C



(d) Item D



(e) Item E

Figure 1: Demo figure grouped into two labeled rows.

9 Tables — Multirow, Multicolumn & Color

9.1 Multirow (vertical merge) + multicolumn (horizontal merge)

```
\begin{tabular}{l c c c}
\toprule
\multirow{2}{*}{\textbf{Family}} & \multicolumn{2}{c}{\textbf{Transform}} & \multirow{2}{*}{\textbf{Time}} \\
\cmidrule(lr){2-3}
& Time & Laplace & \\
\midrule
\multirow{2}{*}{Algebraic} & $1$ & $1/s$ & $s>0$ \\
& $t$ & $1/s^2$ & $s>0$ \\
\bottomrule
\end{tabular}
```

Only the *first* row of a merged group writes `\multirow{n}{*}{...}`; the following rows in that group start their line with a bare `&` (empty cell, since the merged content already occupies that visual space).

9.2 Colored tables (needs xcolor with the [table] option)

```
\usepackage[table]{xcolor} % must use this exact option in the preamble
...
\begin{tabular}{l c c}
\rowcolor{headblue}
\textbf{Planet} & \textbf{a (AU)} & \textbf{e} \\
Jupiter & 5.204 &  $4.89 \times 10^{-2}$  \\
\rowcolor{rowgray}
Saturn & 9.583 &  $5.65 \times 10^{-2}$  \\
Uranus & \cellcolor{yellow!30}19.191 &  $4.72 \times 10^{-2}$  \\
\end{tabular}
```

`\rowcolor{color}` shades an entire row (placed right before that row's content); `\cellcolor{color}` shades a single cell only. Renders as:

Planet	a (AU)	e
Jupiter	5.204	4.89×10^{-2}
Saturn	9.583	5.65×10^{-2}
Uranus	19.191	4.72×10^{-2}

9.3 Putting it all together: colored + multirow + multicolumn + math

```
\begin{table}[h]
\centering
\caption{Orbital and relativistic parameters.}
\label{tab:orbits}
\begin{tabular}{l c c c c}
\toprule
\rowcolor{headblue}
\multirow{2}{*}{\textbf{Planet}} & \multicolumn{2}{c}{\textbf{Orbital}} & \multicolumn{2}{c}{\textbf{Relativistic}} \\
\rowcolor{headblue}
& $a$ (AU) & $e$ & $\varepsilon$ & $\beta = \sqrt{\varepsilon}$ \\
\midrule
Jupiter & 5.204 &  $4.89 \times 10^{-2}$  &  $1.90 \times 10^{-9}$  &  $4.36 \times 10^{-5}$  \\
\rowcolor{rowgray}
Saturn & 9.583 &  $5.65 \times 10^{-2}$  &  $1.03 \times 10^{-9}$  &  $3.21 \times 10^{-5}$  \\
\bottomrule
\end{tabular}
\end{table}
```

Renders as:

Planet	Orbital		Relativistic	
	a (AU)	e	ε	$\beta = \sqrt{\varepsilon}$
Jupiter	5.204	4.89×10^{-2}	1.90×10^{-9}	4.36×10^{-5}
Saturn	9.583	5.65×10^{-2}	1.03×10^{-9}	3.21×10^{-5}

10 Theorem, Proposition & Proof

Needs `amsthm`. Declare the environment once in the preamble:

```
\newtheorem{proposition}{Proposition}[section] % numbers as 3.1, 3.2, ... within Sec.3
```

Then use it, followed by a `proof` block (comes free with `amsthm`, ends with an automatic ■ / □ QED mark):

```
\begin{proposition}[Weak-field circular orbit]
  \label{prop:weakfield}
  For a circular orbit of radius  $r$ ,
  \[ \left(\frac{v}{c}\right)^2 = \frac{GM}{rc^2} = \frac{r_s}{2r}. \]
\end{proposition}

\begin{proof}
  The circular-orbit balance  $v^2/r = GM/r^2$  gives  $v^2 = GM/r$ .
  Divide by  $c^2$  and use  $r_s = 2GM/c^2$ .
\end{proof}

See Proposition~\ref{prop:weakfield}.
```

Renders as:

Proposition 10.1 (Weak-field circular orbit). *For a circular orbit of radius r ,*

$$\left(\frac{v}{c}\right)^2 = \frac{GM}{rc^2} = \frac{r_s}{2r}.$$

Proof. The circular-orbit balance $v^2/r = GM/r^2$ gives $v^2 = GM/r$. Divide by c^2 and use $r_s = 2GM/c^2$. □

See Proposition 10.1. (the number 10.1 was generated automatically — never type it by hand.)

11 Footnotes

```
Sizes and distances are schematic.\footnote{Sizes and distances are schematic.}
```

Glue `\footnote{...}` directly onto the word it annotates — no space, no new paragraph.¹ Never type the footnote marker number yourself.

12 Hyperlinks

```
See \href{https://science.nasa.gov/solar-system/}{NASA Solar System Exploration}.
Or show a bare URL: \url{https://science.nasa.gov}
```

Renders as: See [NASA Solar System Exploration](https://science.nasa.gov/solar-system/). Needs `hyperref`, loaded *last* in the preamble to avoid option clashes with other packages.

¹Exactly like this one. LaTeX numbers footnotes automatically and continues counting across the *whole document*, not per page.

13 Citations & Bibliography

Fastest, safest option under time pressure — no extra compiler passes needed:

```
In text: The Solar System formed from a molecular cloud \cite{kepler1609}.
```

At the end of the document:

```
\begin{thebibliography}{9}
  \bibitem{kepler1609} J. Kepler, \textit{Astronomia Nova}. Heidelberg, 1609.
  \bibitem{nasa2024} National Aeronautics and Space Administration,
    ‘‘Orbits and Kepler’s Laws,’’ 2024.
\end{thebibliography}
```

`\cite{key}` prints as a bracketed number, e.g. [1], matched to the `\bibitem{key}` with the same key. Organization-authored sources (like NASA) are typed as an ordinary `\bibitem` — there is no special syntax for a non-person author.

If a `.bib` file is explicitly provided and BibTeX use is clearly expected instead, the alternative is:

```
\bibliographystyle{plain}
\bibliography{laplace}      % without the .bib extension
```

This needs an extra `bibtex` pass between two `pdflatex` passes (`pdflatex` → `bibtex` → `pdflatex` → `pdflatex`). Know the manual `thebibliography` method as your fallback if that fails on an unfamiliar lab machine.

14 Code Listings

```
\usepackage{listings}
\lstset{language=Python, basicstyle=\ttfamily\small, numbers=left, frame=single}

\lstinputlisting[caption={Computing RC step response.}, label={lst:rc}]{rc_response.py}
% or, to type code directly instead of pulling in a file:
\begin{lstlisting}[caption={Inline example}]
def square(x):
    return x * x
\end{lstlisting}
```

Prefer `\lstinputlisting` whenever an external file is provided — it eliminates transcription typos entirely.

15 Compilation Checklist & Common Errors

15.1 Compile order

Document contains	Run
Only text/equations/figures	<code>pdflatex</code> once
Cross-references (<code>\ref</code> , TOC)	<code>pdflatex</code> $\times 2$
Manual <code>\thebibliography</code>	<code>pdflatex</code> $\times 2$ (same as above)
BibTeX (<code>.bib</code> file)	<code>pdflatex</code> $\rightarrow$ <code>bibtex</code> $\rightarrow$ <code>pdflatex</code> $\times 2$

?? in place of a number, or an “undefined reference” warning, after only *one* compile is completely normal — it disappears on the second pass.

15.2 Common error messages

Error	Typical cause	Fix
Undefined control sequence	misspelled command, or its package isn’t loaded	check spelling; add the package
Missing \$ inserted	a math symbol (<code>_, \frac</code> , etc.) used outside math mode	wrap it in <code>\$. . . \$</code>
Misplaced alignment tab character &	a bare <code>&</code> used outside <code>tabular/align</code>	escape it as <code>\&</code> in plain text
Runaway argument	a brace <code>{</code> was never closed	count braces; add the missing <code>}</code>
Too many <code>}</code> ’s	one extra closing brace	remove the extra <code>}</code>
File not found	wrong filename/path, or extension typed when not needed	check exact filename (case-sensitive)
Undefined reference	label doesn’t exist yet, or you compiled only once	compile again; check the <code>\label</code> key matches <code>\ref</code> exactly

15.3 Final checklist before you submit

- Compiled at least twice (three times if using BibTeX) so all numbers resolve.
- No stray ?? left in the PDF.
- Every `\label` has a matching `\ref/\eqref` being used somewhere, and vice versa.
- Exact page/float placement does *not* need to match the original — don’t burn time fighting `[h]` vs. `[htbp]`.